

Learning to Replenish and Audit under Limited Inventory Traceability

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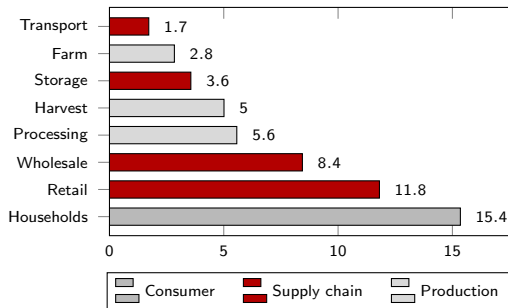
Outline

- 1 Motivation
- 2 Model
- 3 Algorithm
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Where Is Food Lost? Evidence Across the Chain



Three clusters of stages

- **Consumer:** Household
- **Production:** Farm, Harvest, Processing
- **Supply chain:** Retail, Wholesale, Storage, Transportation

Source: FAO Food Loss and Waste Database — the UN specialized agency leading global efforts against hunger, working in over 130 of its 194 members to advance food security.

Motivation

Summed across stages, the **supply chain** (retail, wholesale, storage, transportation) accounts for the **largest combined share** of average food loss. This paper focuses on **optimizing supply chain decisions**—inventory planning and management—with **shrinkage in consideration**.

Motivating Setting: A Growing Online Grocer

- Small online grocer in **Eastern Canada** connecting local farmers and specialty producers with consumers.
- Assortment grew from a few hundred SKUs to more than **10,000** in under three years.
- Moved into a rented warehouse to consolidate inbound shipments and fulfill orders.
- Item-level visibility became difficult to maintain with the firm's **limited data infrastructure**.

Why this setting is hard

Unrecorded spoilage, breakage, theft, damage, and misplacement all create the **same information problem**: the operational estimate can exceed what can actually be located and sold.

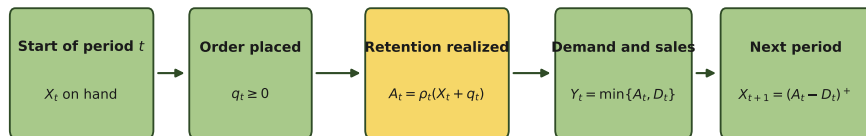
Research Questions

How should a retailer **replenish** from an imperfect state estimate, and when is a physical count **worth its cost**?

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A Periodic-Review Model with Hidden Shrinkage



Hidden: X_t, A_t, ρ_t, L_t | Observed: order q_t , sales Y_t , stockout Z_t

- Retention factor $\rho_t \in [0, 1]$: the fraction $1 - \rho_t$ is rendered unavailable by unrecorded loss.
- Sales are censored by usable inventory: $Y_t = \min\{A_t, D_t\}$; lost demand $L_t = (D_t - A_t)^+$.
- Cost $C_t = hX_{t+1} + bL_t$: holding cost h and lost-sales penalty b .

Physical Audits Reveal Retention Exactly

- An audit may start only at a **known-zero stockout boundary**: the learner places a fixed audit-start order s_{aud} and, at period end, physically counts ending inventory X_{t+1} .
- Because the audit begins from zero true inventory,

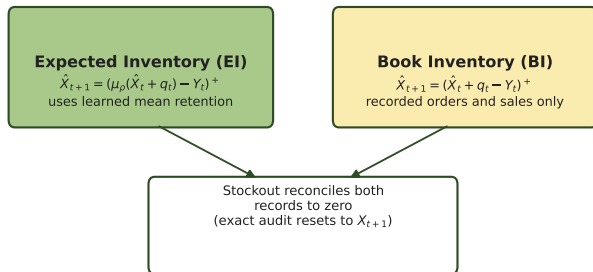
$$\rho_t = \frac{Y_t + X_{t+1}}{s_{\text{aud}}}.$$

- ⇒ The audit reveals the current retention realization **exactly**, whether or not the audit period stocks out.
- Each count costs a known fee κ_{aud} .

The trade-off

Audits purchase **exact information** at a direct fee; ordinary operation is **free but only partially informative** (censored by stockouts).

Two Ways to Track Inventory: EI vs. BI



- Following a nonstockout, **BI** replenishes observed sales; **EI** also adds $(1 - \mu_\rho)S$, an adjustment for average shrinkage.
- In either mode, a reliable stockout reconciles the tracker to zero; an audit reconciles it **exactly**.
- BI needs no estimate of μ_ρ to run—but the learner still needs to know *which* rule is cheaper.

The Benchmark and the Regret Objective

Non-auditing benchmark

$$(c_{\mathcal{U}}^*, S_{\mathcal{U}}^*) \in \arg \min_{\substack{c \in \{\text{EI}, \text{BI}\} \\ S \in [0, \bar{S}]}} g_c(S), \quad g_{\mathcal{U}}^* := g_{c_{\mathcal{U}}^*}(S_{\mathcal{U}}^*).$$

The benchmark knows the mean retention factor, the better rule, and the best target—but never audits.

Regret

$$R_T := \mathbb{E} \left[\sum_{t=1}^T C_t^{\text{alg}} \right] - T g_{\mathcal{U}}^*.$$

What makes this hard

The learner must determine **which rule** (EI or BI) is cheaper, **which target** S is best, and **when to pay** for an exact observation—all from censored, delayed signals.

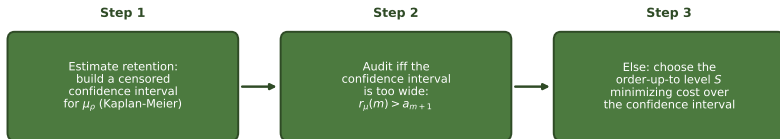
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KM-Audit at 30,000 Feet

KM-Audit — Kaplan–Meier Audit

At every stockout boundary, KM-Audit **estimates retention**, checks **whether to audit**, and **picks the best order-up-to level**.

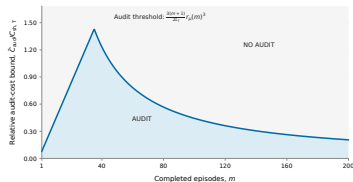


- **Step 1:** Pool exact (audit) and right-censored (routine) observations of ρ_t into a Kaplan–Meier confidence interval for μ_p .
- **Step 2:** Audit only if that interval is still **too wide**, $r_\mu(m) > a_{m+1}$.
- **Step 3:** Choose the order-up-to level S (and EI/BI rule) **minimizing cost**, evaluated optimistically within the interval.

Adaptive Audit Rule: A Cube-Root Balance

Cube-root threshold

$$a_m := \min \left\{ 1, \vartheta_T \left(\frac{L_T}{m} \right)^{1/3} \right\}$$
$$\vartheta_T := \left(\frac{2\bar{C}_{\text{aud}}}{3C_{\text{sh},T}} \right)^{1/3}$$



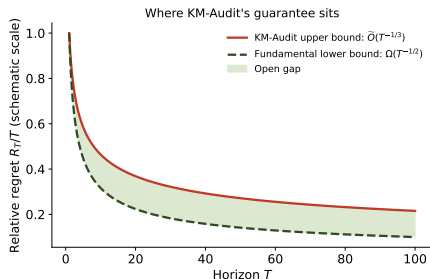
Audit rule

Audit at episode $m+1$ iff $r_\mu(m) > a_{m+1}$.

Intuition: balancing two costs at radius b_T

- $r_\mu(m)$ is a **margin of error** on μ_ρ (like a poll's); n audits shrink it like $1/\sqrt{n}$ —reaching radius b_T needs $n \sim L_T/b_T^2$ audits, so **halving** the margin costs **4×** the audits.
 - Leftover uncertainty is a **running tax**: $\approx T$ episodes at radius $\approx b_T$ each pay $\propto b_T$, for a cumulative tax $\propto T b_T$.
- $\Rightarrow \text{Cost}(b_T) \approx \underbrace{\bar{C}_{\text{aud}} L_T / b_T^2}_{\text{audits}} + \underbrace{C_{\text{sh},T} T b_T}_{\text{tax}}$: equating the two terms gives $b_T \propto T^{-1/3}$ and
- $\text{Cost} \propto T^{2/3}$.

Regret Analysis



Where things stand

- Upper bound:
 $R_T/T = \tilde{O}(T^{-1/3})$, achieved by KM-Audit.
- Lower bound:
 $R_T/T = \Omega(T^{-1/2})$, from retention-law uncertainty alone.
- Both vanish as $T \rightarrow \infty$ —the **minimax rate** at which they do remains open.

Managerial reading

Delayed, partially censored operational signals (like shrinkage) create **genuinely new** exploration–exploitation problems, even when the underlying control model is familiar.

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Retailer Data and Evaluation Design

Data

Transaction-level fulfilled-order data from an online grocer. Learning policies are **calibrated on 2023**, tested on the **2024 fulfilled-order sequence** (319 days, Jan 1–Nov 14).

Cohort

101 SKUs across 9 categories (bakery, beverages, dairy, fish/seafood, meat, pantry, prepared meals, produce, vegetarian), after activity and data-quality screens.

Shrinkage scenario

One external, category-calibrated food-loss scenario generates hidden inventory paths (not observed at the retailer).

- Counted-state protocol: KM-Audit uses **2** fixed counts, an 11-period high-stock block, then 306 ordinary periods.
- Performance is measured relative to a **restricted hindsight fixed-stock** benchmark (percentage cost difference).

Benchmark Policies

Audit-assisted

- **Stock-up + audits:** 8 counts, then stock generously and learn demand empirically.
- **ETC(β):** $\beta \in \{0.4, 0.5, 0.6\}$; count-then-explore, then commit.

No audits

- **Lean stock:** $\hat{\mu} = 1$; rolling KM demand estimate; never counts.
- **Shrinkage-blind AIM:** stochastic-subgradient target update [Huh and Rusmevichientong, 2009]; ignores shrinkage entirely.

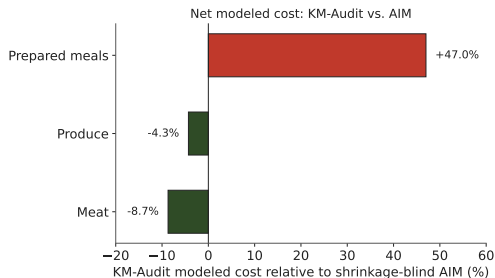
- All policies use a plug-in newsvendor decision $\hat{S}$ from estimated demand and retention.
- Any physical count's fee is charged to **total modeled cost**.
- **KM-Audit** in the experiments uses only its **EI** rule with a predeclared count budget—not the full adaptive audit rule or EI/BI selection.

Net Modeled Cost: KM-Audit vs. Shrinkage-Blind AIM

Main takeaway

KM-Audit's modeled cost advantage is

category-dependent: favorable in meat and produce, unfavorable in prepared meals.

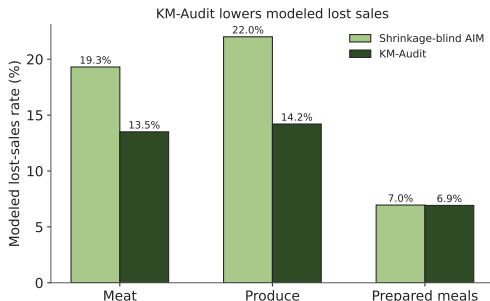


- **Meat:** KM-Audit is **8.7% below** AIM in modeled cost (163.2% vs. 188.4% above the fixed-stock benchmark).
- **Produce:** KM-Audit is **4.3% below** AIM.
- **Prepared meals:** KM-Audit is **47.0% above** AIM—extra counting and inventory exposure without a matching service gain.

Better State Information Improves Modeled Service

Main takeaway

KM-Audit **lowers modeled lost sales** relative to AIM in every displayed category.

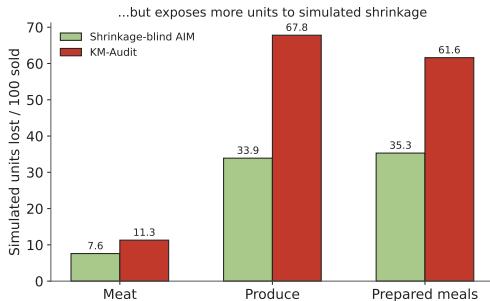


- Meat: lost-sales rate falls from **19.3%** (AIM) to **13.5%** (KM-Audit).
- Produce: falls from **22.0%** to **14.2%**.
- Prepared meals: essentially unchanged (**6.95%** vs. **6.92%**)—little to gain here.

...But Higher Service Means More Exposure to Shrinkage

Main takeaway

Carrying more usable inventory to serve demand also **exposes more units** to the simulated shrinkage process.



The service-loss trade-off

KM-Audit has the **highest** simulated unit loss in all nine categories, yet the **lowest** modeled lost-sales rate in eight. Whether this trade is worthwhile is category-specific.

No Universal Winner: Category Heterogeneity

- Among the 7 categories with ≥ 5 SKUs, KM-Audit ranks **first in 3**, second in dairy, third in bakery, fish/seafood, and prepared meals.
- KM's modeled cost **advantage** is not statistically resolved in any single category; its **disadvantage** is resolved in bakery, fish/seafood, and prepared meals.
- **Meat** is the clearest case: a statistically resolved service gain with a modest increase in simulated loss.
- **Prepared meals** should not receive a count budget on category label alone—no real service gain, large increase in simulated loss.

Practical implication

Validate a category's **actual** shrinkage process before committing to an audit-assisted pilot.

Summary

- Unrecorded shrinkage turns replenishment into both a **control** problem and an **information** problem.
- KM-Audit pools exact (audit) and right-censored (routine) retention observations into one Kaplan–Meier confidence set.
- It jointly learns the base-stock target, chooses between EI and BI, and decides **when a count is worth its cost**.
- We establish $\tilde{O}(\tau^{2/3})$ regret against the better non-auditing benchmark, and an $\Omega(\sqrt{\tau})$ floor from retention-law uncertainty alone.
- On real retailer data, KM-Audit's benefit is **scenario- and category-conditional**, not universal.

Thank you for your attention.

Questions and comments welcome.

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Model Assumptions

- **Bounded demand:** $0 \leq D_t \leq \bar{D}$ a.s. for a known constant $\bar{D}$.
- **Compact mean-retention interval:** $\mu_\rho \in [\underline{\mu}, \bar{\mu}] \subset (0, 1)$.
- **Positive stockout probability:** $\mathbb{P}(Z_t = 1 \mid \mathcal{F}_t^-) \geq p_0 > 0$ uniformly over every policy and audit episode in the class considered—gives geometric control of cycle lengths.
- **Bounded demand density:** $\mathbb{P}(D_t \in I) \leq \bar{f}_D |I|$ for any interval I .

Role of these conditions

Boundedness conditions make cycle statistics well behaved; the stockout-regeneration condition gives **uniform geometric control** over cycle lengths across every policy considered, comparator and learner alike.

The KM-Audit Algorithm (Pseudocode Overview)

- ❶ **Warm start:** take one audit before adaptive decisions begin.
- ❷ **At each episode boundary m :**
 - ❶ Update the pooled mean-retention confidence set $\mathcal{C}_\mu(m)$, estimate $\hat{\mu}_m$, radius $r_\mu(m)$.
 - ❷ Update pseudo-cost estimates and LCB/UCB for all active policy arms; eliminate dominated arms.
 - ❸ If $r_\mu(m) > a_{m+1}$: **audit**. Else: play the optimistic (lowest-LCB) arm, or an unsampled arm if one remains.
- ❸ **At the horizon:** return the best sampled arm's rule and target $(\hat{c}_T, \hat{S}_T)$.

Design principle

Policy-arm elimination and the audit decision share the **same** confidence machinery and episode clock—so shrinkage learning and policy learning inform each other continuously.

The Hard Instance Behind the $\Omega(\sqrt{T})$ Lower Bound

Construction

Fix $h = 1$, $b = 10$, $\bar{S} = 3$, known demand $D_t \sim \text{Unif}[1, 1.2]$, and mean-retention interval $[3/16, 9/40]$. Consider two retention laws $P_{\Delta}^+, P_{\Delta}^-$ on $\{0, 0.1, 0.6\}$ with means $\mu_{\pm} = 33/160 \pm \Delta/2$.

- Even with an **oracle** revealing true inventory, demand, and retention every period, one period supplies only $O(\Delta^2)$ information distinguishing the two laws.
- Choosing $\Delta_T = 1/(8\sqrt{40T})$ makes the environments hard to distinguish over T periods, while the wrong choice costs $\Theta(\Delta_T)$ per period.

$\Rightarrow T\Delta_T = \Theta(\sqrt{T})$ regret is unavoidable.

Interpretation

The extra difficulty from shrinkage is **fundamental**, not an artifact of KM-Audit's design.

Retailer Cohort: Construction and Screens

Predeclared screens (2023)

Maps to product/category masters; ≥ 20 positive-order days and ≥ 30 fulfilled units; first sale by March 1; active through Nov. 1; no zero-order spell > 60 days $\Rightarrow$ **112 SKUs**.

Further screens

At least one positive 2024 order $\Rightarrow$ 104 SKUs. Exclude 3 SKUs with nonpositive/unavailable margin proxy $\Rightarrow$ **101 SKUs**.

Bakery	Beverages	Dairy	Fish/Seafood	Meat	Pantry
18	2	11	10	21	9
Prepared meals		Produce	Vegetarian		
16		12	2		

References I

Woonghee Tim Huh and Paat Rusmevichientong. A nonparametric asymptotic analysis of inventory planning with censored demand. *Mathematics of Operations Research*, 34(1):103–123, 2009.

When Book Inventory Lies

- Classical inventory theory assumes the physical state is **observable** at each decision epoch.
- In practice, **book inventory**—what records say should be there—can persistently overstate what is actually sellable.
- Theft, damage, misplacement, unrecorded disposal, and spoilage create **inventory shrinkage**: unrecorded loss that is invisible until a stockout or a physical count.

Sources of shrinkage

- Theft
- Damage
- Misplacement
- Unrecorded disposal
- Spoilage

Bottom line

A record describes what **should** be available; replenishment depends on what can actually be **found and sold**.

Physical Counts Restore Visibility — But Are Not Free

What routine operation reveals

- A **stockout** reveals sellable inventory has hit zero—but only after unserved demand.
- Orders and sales are recorded; the true on-hand quantity is not.

What a physical count reveals

- An exact SKU-level count restores visibility.
- Counting every SKU before every order is feasible for a small assortment, **prohibitive** for thousands of items.

Research premise

The retailer must substitute between **exact information** purchased through a targeted count and the **imperfect signals** generated by routine sales and stockouts.

Motivating Setting: A Growing Online Grocer

- Small online grocer in **Eastern Canada** connecting local farmers and specialty producers with consumers.
- Assortment grew from a few hundred SKUs to more than **10,000** in under three years.
- Moved into a rented warehouse to consolidate inbound shipments and fulfill orders.
- Item-level visibility became difficult to maintain with the firm's **limited data infrastructure**.

Why this setting is hard

Unrecorded spoilage, breakage, theft, damage, and misplacement all create the **same information problem**: the operational estimate can exceed what can actually be located and sold.

The managerial question

How should a retailer **replenish** from an imperfect state estimate, and when is a physical count **worth its cost**?

Research Questions

Core research contribution

We study a periodic-review lost-sales system with hidden shrinkage and develop **KM-Audit**, a policy that jointly learns replenishment, the record-updating rule, and when to audit.

- **Record-updating:** Should the internal record track **expected** shrinkage (EI) or reset only at reliable **stockouts** (BI)?
- **Replenishment:** What base-stock target should the retailer use given a hidden physical state?
- **Information acquisition:** When does a paid physical count **pay for itself** relative to the performance loss from uncertainty?
- **Guarantees:** How quickly can a learning policy approach the best non-auditing benchmark, and what is **unavoidable**?

Physical Audits Reveal Retention Exactly

- An audit may start only at a **known-zero stockout boundary**: the learner places a fixed audit-start order s_{aud} and, at period end, physically counts ending inventory X_{t+1} .
- Because the audit begins from zero true inventory,

$$\rho_t = \frac{Y_t + X_{t+1}}{s_{\text{aud}}}.$$

- ⇒ The audit reveals the current retention realization **exactly**, whether or not the audit period stocks out.
- Each count costs a known fee κ_{aud} ; total algorithm cost is $C_t^{\text{alg}} = C_t + \kappa_{\text{aud}} J_t^{\text{aud}}$.

The trade-off

Audits purchase **exact information** at a direct fee; ordinary operation is **free but only partially informative** (censored by stockouts).

The Benchmark and the Regret Objective

Non-auditing benchmark

$$(c_{\mathcal{U}}^*, S_{\mathcal{U}}^*) \in \arg \min_{\substack{c \in \{\text{EI}, \text{BI}\} \\ S \in [0, \bar{S}]}} g_c(S), \quad g_{\mathcal{U}}^* := g_{c_{\mathcal{U}}^*}(S_{\mathcal{U}}^*).$$

The benchmark knows the mean retention factor, the better rule, and the best target—but never audits.

Regret

$$R_T := \mathbb{E} \left[\sum_{t=1}^T C_t^{\text{alg}} \right] - T g_{\mathcal{U}}^*.$$

What makes this hard

The learner must determine **which rule** (EI or BI) is cheaper, **which target** S is best, and **when to pay** for an exact observation—all from censored, delayed signals.

Stockout Cycles Reveal Observable Pseudo-Costs

- True period cost C_t is **hidden**: lost demand and true inventory are unobserved.
- A completed stockout cycle reveals two observable aggregates: total sales V_k and total loss $W_k = Q_k - V_k$ (orders minus sales).

Proposition 1 (Observable pseudo-cost)

For $\alpha(\mu) := h\mu/(1 - \mu)$, the long-run average cost satisfies

$$g(S; \nu) = b \mathbb{E}[D_t] + \alpha(\mu_\rho) w(S; \nu) - (h + b) v(S; \nu),$$

where v, w are the per-period rates of V_k, W_k . Minimizing $g_c(S)$ is equivalent to minimizing this **pseudo-cost rate**.

Why this matters

The learning problem separates into estimating μ_ρ and comparing **policy arms** on their pseudo-cost.

Episode Structure: Decision Cycles vs. Audit Episodes

Decision cycle

- Selects a policy arm $(i, c) \in \mathcal{S}_M \times \{\text{EI}, \text{BI}\}$.
- Runs until the next stockout.
- Yields a **censored** retention observation and cycle statistics (Λ, Q, V, W) .

Audit episode

- Orders the fixed s_{aud} .
- Counts ending inventory exactly.
- Yields an **exact** retention observation; may be followed by a bounded recovery tail.

Discretization

Base-stock targets are discretized to $\mathcal{S}_M = \{s_1, \dots, s_M\}$, giving $2M$ policy arms—a **learning device**, not a restriction on the (continuous) benchmark.

Retention Learning from Pooled Censored Observations

- Every episode start contributes one observation to a pooled sample $\mathcal{D}_p$.
- **Decision cycles** contribute **right-censored** observations of ρ_t (the tracker recursion substitutes an unobserved draw).
- **Audits** contribute **uncensored** observations via the exact identity $\rho_t = (Y_t + X_{t+1})/s_{\text{aud}}$.

Three complementary confidence intervals for μ_p

- A distribution-free **bracketing** interval.
- An **audit-only Hoeffding** interval (fallback when risk sets are thin).
- A **Kaplan–Meier** interval that adapts to how informative ordinary operation has been.

Key idea

Informative unaudited episodes can shrink the confidence radius **without paying for a count**.

Policy-Arm Selection: Optimism with Elimination

Optimistic selection

Among non-eliminated arms, play the one with the smallest **lower confidence bound** (LCB) on pseudo-cost.

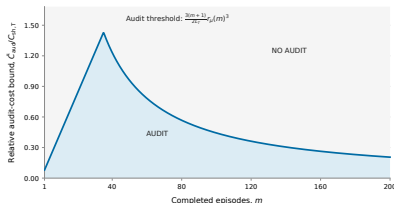
$$\mathcal{A} \leftarrow \left\{ (i, c) \in \mathcal{A} : \text{LCB}_{i,c}(m) \leq \min_{(j,d) \in \mathcal{A}} \text{UCB}_{j,d}(m) \right\}$$

- Arms are **eliminated** once their LCB exceeds the best UCB among active competitors.
- Evaluation uses the current confidence set for μ_ρ —so arm comparison and shrinkage learning are **coupled**.
- The two inventory-tracking modes (EI, BI) compete on **equal footing** within the same elimination rule.

Adaptive Audit Rule

Cube-root threshold

$$a_m := \min \left\{ 1, \vartheta_T \left(\frac{L_T}{m} \right)^{1/3} \right\}$$
$$\vartheta_T := \left(\frac{2\bar{C}_{\text{aud}}}{3C_{\text{sh},T}} \right)^{1/3}$$



Audit rule

Audit at episode $m+1$ iff $r_\mu(m) > a_{m+1}$.

- $\bar{C}_{\text{aud}}$: uniform bound on expected audit-episode cost. $C_{\text{sh},T}$: converts a unit of confidence radius into a cost bound.
- The threshold **balances** the bounded cost of counting against the performance loss from unresolved retention uncertainty.

Main Regret Upper Bound

Theorem 1 (Informal)

Under the model's boundedness and regeneration assumptions, KM-Audit's regret satisfies

$$R_T \leq \underbrace{C_{\text{bd}} + \bar{C}_{\text{aud}} \mathbb{E}[A_T^{\text{init}}] + C_{\text{sh},T} \mathbb{E}[\mathcal{U}_T^\mu]}_{\text{audit \& shrinkage-uncertainty terms}} + \underbrace{\frac{2\bar{c}_{\text{per}}}{p_0}(2M) + \frac{2}{p_0} \mathbb{E}[\mathcal{W}_T^{\text{stat}}] + T\varepsilon_M}_{\text{policy-arm learning \& grid error}}.$$

- Choosing the grid size $M_T = \lceil T^{1/3} \rceil$ balances all terms:

$$R_T = \tilde{O}(T^{2/3}).$$

- If ordinary observations alone keep $\mathbb{E}[\mathcal{U}_T^\mu] = \tilde{O}(\sqrt{T})$ with $O(1)$ audits, shrinkage learning contributes only $\tilde{O}(\sqrt{T})$.

Why $\tilde{O}(T^{2/3})$? A Cube-Root Balance

Buying precision is expensive

$r_\mu(m)$: margin of error on μ_ρ , like a poll's.

- n exact (audit) observations shrink it like $1/\sqrt{n}$.

⇒ Reaching radius b_T needs $n \sim L_T/b_T^2$ audits: **halving** the margin costs **4×** the audits.

Leftover uncertainty is a running tax

Each episode starting at radius $\approx b_T$ pays $\propto b_T$ (via $C_{\text{sh},T}$): a blurred tracker and cost comparison.

- Over $\approx T$ episodes, cumulative radius $\sum_m r_\mu(m)$ grows like $T \cdot b_T$.
- **Larger** $b_T \Rightarrow$ bigger running tax.

Balance: total cost as a function of b_T

$$\text{Cost}(b_T) \approx \underbrace{\bar{C}_{\text{aud}} \frac{L_T}{b_T^2}}_{\text{audits}} + \underbrace{C_{\text{sh},T} T b_T}_{\text{residual tax}} \implies b_T \propto T^{-1/3}, \quad \text{Cost} \propto T^{2/3}.$$

Smaller b_T : fewer, cheaper audits but higher tax; setting the two terms to the **same order** picks the best trade-off—and both then scale as $T^{2/3}$.

A Fundamental $\Omega(\sqrt{T})$ Lower Bound

Theorem 2 (Informal)

*Even under a substantial **oracle relaxation**—true inventory, demand, and retention are revealed each period, and the learner may use arbitrary state-dependent orders—*

$$\inf_{\pi} \sup_{M \in \mathcal{M}_{\text{sh}}} R_T^{\pi}(M) \geq \Omega(\sqrt{T}).$$

- Demand is **fixed and known**; all uncertainty is in the retention law.
- Two retention laws differing by $\Delta_T \propto 1/\sqrt{T}$ are statistically hard to distinguish, yet choosing the wrong side incurs $\Theta(\Delta_T)$ loss per period.

Takeaway

Uncertainty about **the retention law itself**—not demand uncertainty or censoring—is enough to force $\Omega(\sqrt{T})$ regret.

Managerial Implications

- ① **Counts inform two decisions:** not only the magnitude of shrinkage, but **whether** to compensate for it at all (EI vs. BI).
- ② **Audit intensity should adapt:** count only while uncertainty remains above a cost-calibrated threshold; informative routine data can **avert** a count.
- ③ **No universal audit policy:** the value of counting is **heterogeneous** across categories and shrinkage regimes.
- ④ **Validate before piloting:** category labels are not a reliable proxy for a SKU's actual shrinkage process.

Who should care

This problem sits at the intersection of **inventory control, information acquisition, and retail operations** under limited traceability.

Research Agenda Going Forward

- **Covariate-driven shrinkage:** let retention vary with location, handling, supplier, batch, or time since receipt.
- **Age-structured extensions:** needed when expiration and FIFO/FEFO allocation matter beyond an aggregate retention factor.
- **Positive lead times:** would couple current audits with future arrivals and weaken the regenerative structure.
- **Multi-item audits:** bundle counts across neighboring SKUs to share information-acquisition cost.

Big picture

Limited traceability need not be all-or-nothing: routine data can determine **when** exact visibility is worth its cost.